



ANN CAPSTONE PROJECT

This project helps you to apply **Python, Numpy, Matplotlib, Tensorflow, Keras, ANN and its variations**. This project will help you in understanding how a proper deep learning code is written and fine tuned.

Modules utilized:

- Numpy
- Pandas
- Keras tuner
- Tensorflow
- Keras

Steps Involved:

1. Download the kaggle dataset which has been assigned to you. Understand the problem statement which is available in the Kaggle Description or Select your own dataset
2. Importing data
 - 2.1. Use pandas or glob whichever you are comfortable with.
3. Data set exploration
 - 3.1. Understanding what data have been given to you
 - 3.2. Perform non visual analysis
 - 3.3. Perform Visual Analysis
 - 3.4. Write down your observations, insights and recommendations(if applicable)

Note 🚩: **Kindly get it reviewed the above steps with the @mentors before proceeding further in the project**

4. Data Preprocessing
 - 4.1. If you find that there is an imbalance in the data set, use techniques like SMOTE, B-SMOTE and ADA-SYN to take care of it.

Note 🚩: **Kindly get it reviewed the above steps with the @mentors before proceeding further in the project**

5. Model Building
 - 5.1. Vanilla ANN (A standard ANN architecture)
6. Fine Tuning: Use Keras tuner to find the best architecture.
7. Compare the performance of ANN v/s M.L. Algorithms (perform hyperparameter tuning if required)

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8. Model Evaluation
 - 8.1. Select appropriate metric according to a classification task
9. Write down your observations for all the steps you perform.
10. Write down a final conclusion after model building and its evaluation.
11. Once the coding part is done write a report on your findings based on the project done by you.

References: Blogs on Data Imbalance

- [Tour of Data Sampling Methods for Imbalanced Classification - MachineLearningMastery.com](#)
- [SMOTE for Imbalanced Classification with Python - MachineLearningMastery.com](#)
- [Class Imbalance, SMOTE, borderline SMOTE, ADASYN | by Dr. Saptarsi Goswami | Towards Data Science](#)

After the completion of the project add it to your profiles such as **LinkedIn, GitHub and CV/Resume** which will put more weightage to your Resume and Digital Profiles.

Presentation, Project VIVA and Code

Walkthroughs:

Based on the project completion you have to give the presentations about what you infer from your business problem followed by a viva. We will schedule a presentation once the Project is submitted. Please contact your Mentors for the Presentation Schedule. Code Walkthroughs will also be conducted in the same way.

 **Click here to [find the PPT Template for the Project Presentation](#)**

Note : Kindly get it reviewed the above steps with the @mentors before presenting

Submission:

After completion of the project Zip the .ipynb notebook and the exported model and upload the zip file with your name and batch number.

Note : For the doubts clarifications you are welcome to join our One to One mentorship session